STAT 7900: Special Topic: Causal Analysis



Instructor

Dhrubajyoti Ghosh

Kennesaw State University

STAT 7900

Recap: the causal question

Descriptive: how do outcomes differ across observed groups?

Predictive: who is likely to have the outcome?

Causal: what would happen if we assigned an intervention?

Specify population, treatment, comparator, outcome, and time horizon.

Example: does Drug A lower four-week blood pressure in eligible adults?

Potential outcomes and the target effect

$Y_i(1)$ and $Y_i(0)$ describe the same person under two treatments.

We observe only

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0).$$

$$ATE = E[Y(1) - Y(0)]$$

in the target population.

$$ATT = E[Y(1) - Y(0) \mid T = 1]$$

among treated people.

$E[Y \mid T = 1] - E[Y \mid T = 0]$ can differ from the ATE.

Four assumptions for causal interpretation

Consistency: treatment received matches the intervention defining $Y(t)$.

Exchangeability:

$$\{Y(1), Y(0)\} \perp T \mid X.$$

Positivity: both treatments occur within relevant X profiles.

No interference: another person's treatment does not change my outcome.

Measured covariate balance alone cannot verify all four assumptions.

Adjustment depends on the causal graph

Role	Example path	For the total effect
Confounder	$T \leftarrow C \rightarrow Y$	Block the backdoor path
Mediator	$T \rightarrow M \rightarrow Y$	Preserve this causal pathway
Collider	$T \rightarrow M \leftarrow U \rightarrow Y$	Avoid opening the closed path

A variable's role depends on the path and the estimand.

Identification through the g-formula

$$E[Y(t)] = E_{\mathrm{x}}\{E[Y \mid T = t, X]\}.$$

Within each X stratum, use comparable observed units to estimate the treatment-specific mean.

Average those means over one common target distribution of X.

The DAG and assumptions justify this replacement of missing counterfactual means.

Identification gives an observed-data formula. Estimation computes it.

g-computation and the treatment coefficient

Fit $m(t,x) = E[Y \mid T = t, X = x]$.

Predict $\hat{m}(1, X_i)$ and $\hat{m}(0, X_i)$ for each person.

Average: $\hat{\tau}_g = n^{-1} \sum_i \{\hat{m}(1, X_i) - \hat{m}(0, X_i)\}$.

With $T \times X$ interaction, the contrast varies with X.

For $m(t,x) = \beta_0 + \beta_T t + \beta_X x + \beta_{TX} tx$, $ATE = \beta_T + \beta_{TX} E[X]$.

Structural causal models

$C := f_C(U_C)$ generates baseline severity.

$T := f_T(C, U_T)$ generates treatment assignment.

$Y := f_Y(C, T, U_Y)$ generates the outcome.

The structural rules state how the system responds to interventions.

A DAG gives the causal structure. An SCM also specifies the mechanisms.

Conditioning, intervention, counterfactual

Operation	Meaning	What stays fixed?
Condition on $T = 1$	Select people who received treatment	Their observed characteristics
$do(T = 1)$	Replace the treatment rule with $T := 1$	Other structural rules
$Y_i(1)$	Outcome for person i under treatment 1	That person's background factors

Intervening changes assignment. Conditioning selects an observed subgroup.

Simulation: a known causal world

C is baseline severity, with

$$C \sim N(0,1).$$

$$P(T = 1 \mid C) = logistic(1.2C).$$

$$Y = 120 + 10C - 6T + \epsilon$$

with independent $\epsilon \sim N(0,25)$.

Higher severity increases treatment probability and blood pressure.

True effect: $Y_i(1) - Y_i(0) = -6\ mmHg$ for every person.

Simulation: generate and compare

```
set.seed(7900)
n <- 5000
C <- rnorm(n)
T <- rbinom(n, 1, plogis(1.2 * C))
eps <- rnorm(n, sd = 5)
Y <- 120 + 10 * C - 6 * T + eps
dat <- data.frame(C, T, Y)
coef(lm(Y ~ T, data = dat))["T"]
coef(lm(Y ~ T + C, data = dat))["T"]
```

Predict the crude and adjusted results before running the code.

Simulation: association versus intervention

Comparison	Approximate result	Reason
Crude observed contrast	$+3.5\ mmHg$	Treated patients have greater severity
Adjustment for C	$-6.0\ mmHg$	Correct outcome model blocks confounding
Same-person interventions	Exactly $-6\ mmHg$	The structural contrast is constant

$$Y_1 = 120 + 10C - 6 + \epsilon \text{ and } Y_0 = 120 + 10C + \epsilon.$$

Randomization and Identification

Under simple random assignment,

$$T \perp \{Y(1), Y(0), X\}.$$

With consistency,

$$E[Y(t)] = E[Y \mid T = t].$$

Therefore

$$ATE = E[Y \mid T = 1] - E[Y \mid T = 0].$$

Randomization balances baseline causes in expectation, not exactly in every sample.

Randomization identifies the assignment effect when outcomes and design support it.

Randomized designs and analysis

Design	Assignment mechanism	Analysis implication
Complete / Bernoulli	Individuals receive random assignments	Respect fixed versus random arm sizes
Blocked / paired	Randomize within strata or pairs	Account for strata or pairs
Cluster	Randomize clinics or other groups	Account for within-cluster dependence

The randomization unit determines how we quantify uncertainty.

Estimation and uncertainty in an RCT

Difference in means: $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$.

Independent-arm SE: $\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}\right)}$

Example: $\hat{\tau} = -5.3$ and $SE = 1.8\ mmHg$.

Approximate 95% CI: $-5.3 \pm 1.96 \times 1.8 = [-8.83, -1.77]$.

A confidence interval measures sampling uncertainty, not the validity of all assumptions.

Assignment, treatment received, and outcome

Z denotes randomized assignment.
T denotes treatment received.

An intention-to-treat comparison preserves the randomized assignment contrast.

Comparing recipients can reintroduce selection when adherence differs.

A causal effect of treatment received needs additional assumptions or design.

Randomizing Z does not automatically make T unconfounded.

Real Data: Job trainings and earnings

LaLonde data: 614 people, 185 treated and 429 comparison units.

Treatment: participation in the job-training program.

Outcome: earnings in 1978, measured in dollars.

Baseline X includes demographics and earnings in 1974 and 1975.

Question: what would earnings be under training versus no training?

Real data: baseline differences

Variable	Treated	Comparison
Age	25.8	28.0
Married	18.9%	51.3%
Black	84.3%	20.3%
1974 earnings	$2,096	$5,619

1978 earnings: $6,349 versus $6,984. Crude difference ≈ −$635.

Real data: standardize model predictions

```
library(MatchIt)
data("lalonde", package = "MatchIt")
d <- lalonde
fit <- lm(re78 ~ treat + age + educ + race + married +
          nodegree + re74 + re75, data = d)
y1 <- predict(fit, transform(d, treat = 1))
y0 <- predict(fit, transform(d, treat = 0))
mean(y1 - y0)
coef(fit)["treat"]
```

These coincide for this additive linear model without treatment interactions.

Real data: what adjustment establishes

Earlier analysis changed a negative crude contrast to a positive adjusted estimate.

Interpretation still depends on no unmeasured confounding given baseline X.

Poor overlap can make predictions rely heavily on extrapolation.

Model estimates and causal credibility must be assessed separately.

The observed data contain one treatment outcome per person, even after fitting a model.

Recap questions and the next step

Why did the simulated crude estimate have the wrong sign?

When does a treatment coefficient equal the g-computation ATE?

Why can treatment received be confounded in a randomized trial?

What checks would make the job-training comparison more credible?

Next: matching and weighting to create a more comparable observational analysis.

STAT 7900 Quiz 4



Propensity scores: the job-training question

Does job training increase earnings in 1978?

$T = 1$ indicates training. $T = 0$ indicates the comparison group.

Baseline X includes demographics and earlier earnings.

Our example uses the 614-person LaLonde dataset throughout.

We need comparable people to stand in for missing counterfactual outcomes.

Why the observed comparison is difficult

Baseline feature	Treated	Comparison
Married	18.9%	51.3%
Black	84.3%	20.3%
1974 earnings	$2,096	$5,619

The groups differ before training begins.

The high-dimensional comparison problem

We want to compare people with similar age, education, and other baseline variables.

Exact agreement on every covariate becomes difficult as X grows.

Earlier earnings are continuous, so exact matches may be rare.

A scalar treatment-assignment summary can help organize comparisons.

The propensity score summarizes assignment using the chosen covariates.

Definition of the propensity score

$$e(x) = P(T = 1 \mid X = x).$$

It is the probability of receiving training given baseline characteristics.

Every participant has a score, whether treated or untreated.

For a given x, the score concerns people sharing that baseline profile.

The observed treatment is 0 or 1. Its probability can lie between them.

Interpreting a score

A participant's estimated score is $\hat{e}(X) = 0.65$.

The fitted model predicts about a 65% training probability for that profile.

The participant can still appear in either treatment arm.

The score does not describe the probability of high earnings or treatment benefit.

A probability of treatment and a causal effect answer different questions.

The score depends on the study sample

The combined sample has 185 treated and 429 comparison participants.

The estimated assignment model describes treatment membership in these study data.

Changing comparison sampling can change the fitted scores.

It does not automatically describe participation probabilities in the whole population.

Specify the study population when interpreting a score.

Which variables belong in the model?

Use baseline covariates justified by the causal question and adjustment graph.

Include measured common causes of training participation and earnings.

Earlier earnings and demographic circumstances motivate the current specification.

Avoid using 1978 earnings, which is the outcome, to predict assignment.

A variable's causal role matters more than its statistical significance.

A logistic treatment model

$$\log \frac{e(X)}{1 - e(X)} = \beta_0 + \beta^{\mathrm{T}} X.$$

The model uses baseline covariates to predict binary treatment membership.

$e(X) = \exp(\eta)/\{1 + \exp(\eta)\}$, where $\eta = \beta_0 + \beta^{\mathrm{T}} X$.

The logistic transform keeps predicted probabilities between 0 and 1.

This regression models treatment assignment, not the outcome.

R: estimate scores in the real dataset

```
library(MatchIt)
data("lalonde", package = "MatchIt")
d <- lalonde
ps_fit <- glm(treat ~ age + educ + race + married +
              nodegree + re74 + re75,
              family = binomial(), data = d)
d$ps <- predict(ps_fit, type = "response")
head(d[c("treat", "age", "educ", "ps")])
```

type = "response" returns probabilities, not log odds.

Fitted scores

Participant	Training	Age	Education	ê(X)
NSW1	1	37	11	0.639
NSW11	1	19	9	0.689
PSID1	0	30	12	0.026
PSID16	0	42	0	0.056

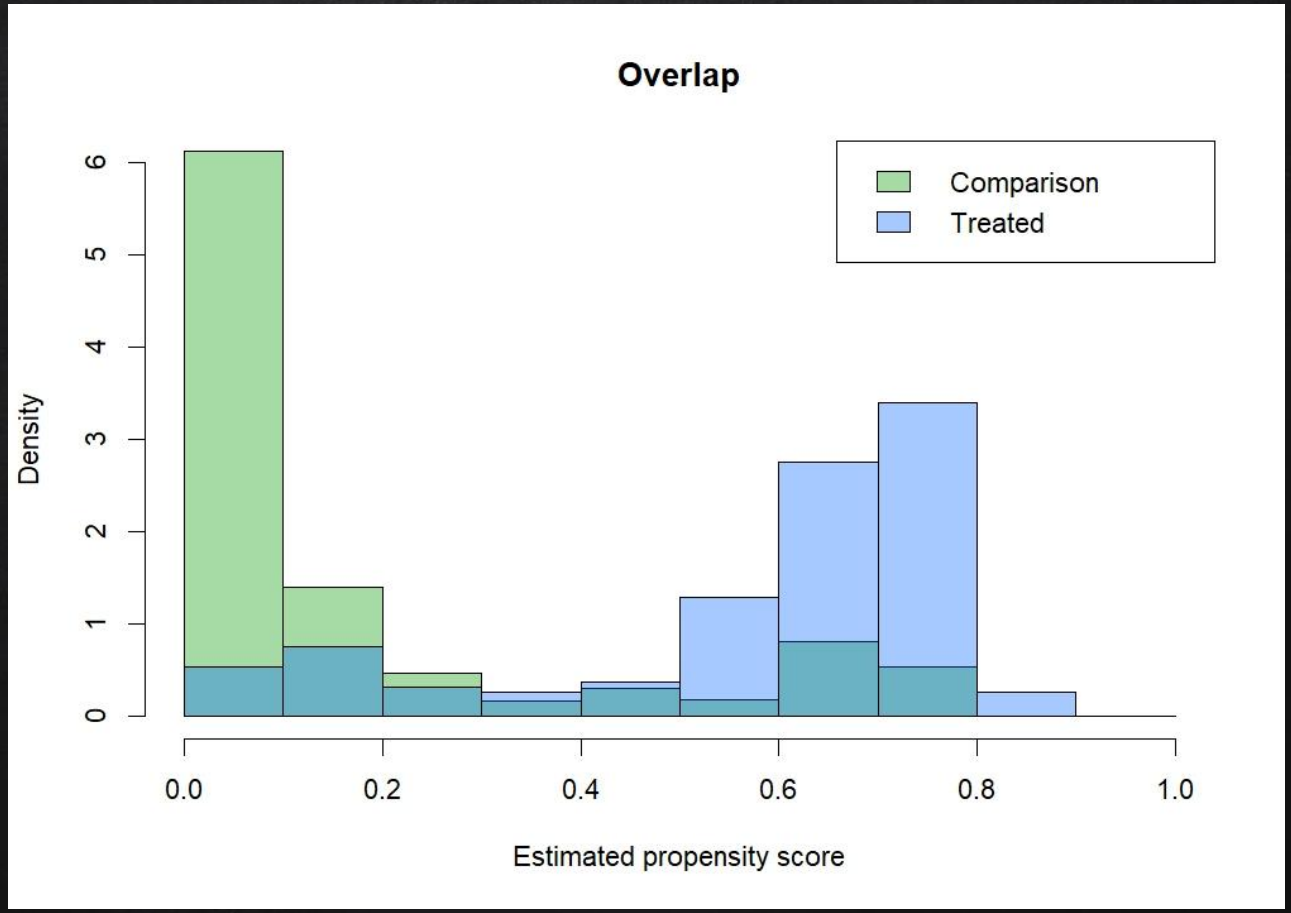
All listed scores also use race, marriage, degree status, and earlier earnings.

Scores differ across observed groups

Group	n	Mean score	Median score	Observed score range
Treated	185	0.577	0.654	0.025–0.853
Comparison	429	0.182	0.076	0.009–0.789

Estimating a propensity score does not itself change either group.

Overlap in the LaLonde data



Positivity and empirical overlap

Population positivity: $0 < e(X) < 1$ for relevant profiles.

Empirical overlap: the sample contains comparable units from both arms.

A logistic model can return $\hat{e} = 0.10$ even for a sparsely supported profile.

Finite scores between 0 and 1 do not establish adequate data support.

A fitted probability cannot create missing counterfactual information.

The propensity score’s balancing property

For the true propensity score:

$$T \perp X \mid e(X).$$

Within a score stratum, treatment assignment does not further select the distribution of X.

This is a population statement, not exact equality between individual profiles.

Estimated scores and finite samples require diagnostic checks.

Comparable score strata can organize a fairer comparison.

Why the balancing property holds

Once $e(X) = s$, every profile in that stratum has treatment probability s.

$$P(T = 1 \mid X, e(X) = s) = s.$$

$$P(T = 1 \mid e(X) = s) = s.$$

Knowing X within the stratum supplies no further assignment information.

Therefore $T \perp X \mid e(X)$.

From balance to causal identification

Assume $\{Y(1), Y(0)\} \perp T \mid X$.

Together with the true balancing score, exchangeability also holds given e(X).

Then $E[Y(t)] = E_{\mathrm{e}}\{E[Y \mid T = t, e(X)]\}$.

Consistency and positivity remain necessary.

The score reduces the adjustment dimension under the causal assumptions.

Similar scores do not mean identical people

Different covariate profiles can produce similar treatment probabilities.

Matching on a score can leave important covariates imbalanced in a sample.

Assess age, prior earnings, and other adjustment covariates individually.

An unmeasured confounder cannot enter this score unless we measure it.

Score similarity is useful, but observed balance still needs checking.

Propensity-score checkpoint

What does a score of 0.65 mean for an untreated participant?

Why do the treated and comparison arms still have different mean scores?

Does a score strictly between 0 and 1 prove positivity?

What would you inspect before interpreting a matched comparison?

Estimating scores is the first step. Comparable groups require a design and balance checks.

Matching: a comparable job-training group

We have estimated each person's probability of receiving training.

Now select untreated participants with similar baseline profiles.

Use their earnings to inform the missing no-training comparison.

The scientific goal remains an effect in a clearly defined population.

Matching uses observed comparison people to support the counterfactual contrast.

Why match when we have g-computation?

Approach	How it constructs the comparison
G-computation	Predict both outcomes using an outcome model
Matching	Select comparable observed treatment and comparison units
Matching + regression	Model outcomes after constructing comparable groups

Matching is an alternative adjustment strategy, not a prerequisite for g-computation.

Benefits and costs of matching

Closer observed comparisons can reduce reliance on outcome extrapolation.

Balance diagnostics make the comparison design visible.

Discarding participants can reduce precision and change the target population.

Matching choices and residual covariate imbalance can affect estimates.

Matching trades sample retention against comparability.

The causal target: ATT

$$ATT = E[Y(1) - Y(0) \mid T = 1].$$

Use treated participants as the focal group.

Find controls to represent their outcomes without training.

Dropping treated participants can change the effect being targeted.

Full ATT and the effect among retained treated participants may differ.

Nearest-neighbor propensity-score matching

For treated participant i, seek a control j with a nearby score.

Distance:

$$d(i,j) = \left|\hat{e}(X_i) - \hat{e}(X_j)\right|.$$

One-to-one matching assigns one eligible control to each retained treated unit.

A greedy algorithm selects each match without optimizing the whole set.

A nearby score is a starting point. Covariate balance determines usefulness.

Actual pairs from the job-training data

Treated ID	Score	Comparison ID	Score	Gap
NSW178	0.853	PSID118	0.789	0.064
NSW95	0.819	PSID226	0.779	0.041
NSW176	0.816	PSID196	0.764	0.052

These are the first three pairs when matching treated scores from largest to smallest.

Matching order and competition for controls

Without replacement, each control can serve only one treated unit.

Early matches remove controls from the pool for later matches.

Different matching orders can produce different pairs.

Record the order so the design is reproducible.

Our example uses descending treated scores: m.order = "largest".

With or without replacement

Choice	Benefit	Cost
Without replacement	Each control appears in one pair	Later treated units may receive weaker matches
With replacement	Good controls can serve multiple treated units	Reuse reduces independent information

A reused control remains one person, even if they appear in several matches.

Calipers: a maximum acceptable distance

A raw score caliper of 0.05 requires $|\hat{e}_i - \hat{e}_j| \leq 0.05$.

A treated person with no eligible control remains unmatched.

Smaller calipers can prevent weak matches but discard more participants.

A caliper can instead be defined on log odds or in SD units.

The number 0.05 has meaning only after specifying the distance scale.

R: the initial matching design

```
library(MatchIt)
data("lalonde", package = "MatchIt")
d <- lalonde
f <- treat ~ age + educ + race + married +
             nodegree + re74 + re75
m0 <- matchit(f, data = d, method = "nearest",
              distance = "glm", estimand = "ATT",
              ratio = 1, replace = FALSE,
              m.order = "largest")
summary(m0)
```

Outcome earnings do not enter the matching formula.

Who remains after initial matching?

Group	Before	After
Treated	185	185
Comparison	429	185
Unused comparison participants	0	244

All treated participants remain, but retention does not establish adequate balance.

Covariate balance: standardized differences

SMD =
(treated mean − comparison mean)/sref.

Use a fixed reference scale when comparing designs.

Here sref is the SD in the original treated group.

Inspect each adjustment covariate. $|SMD| < 0.10$ is a guideline.

A small score difference does not guarantee small covariate differences.

Initial matching improves some covariates

Covariate	Before |SMD|	After |SMD|
Age	0.309	0.072
Married	0.824	0.055
1974 earnings	0.721	0.050
Black indicator	1.757	1.023

Substantial imbalance remains. We should revise the design before relying on its effect estimate.

R: add a raw-score caliper

```
m1 <- matchit(f, data = d, method = "nearest",
              distance = "glm", estimand = "ATT",
              ratio = 1, replace = FALSE,
              m.order = "largest",
              caliper = 0.05, std.caliper = FALSE)
summary(m1)
plot(summary(m1), abs = TRUE)
```

This caliper limits probability differences to 0.05.

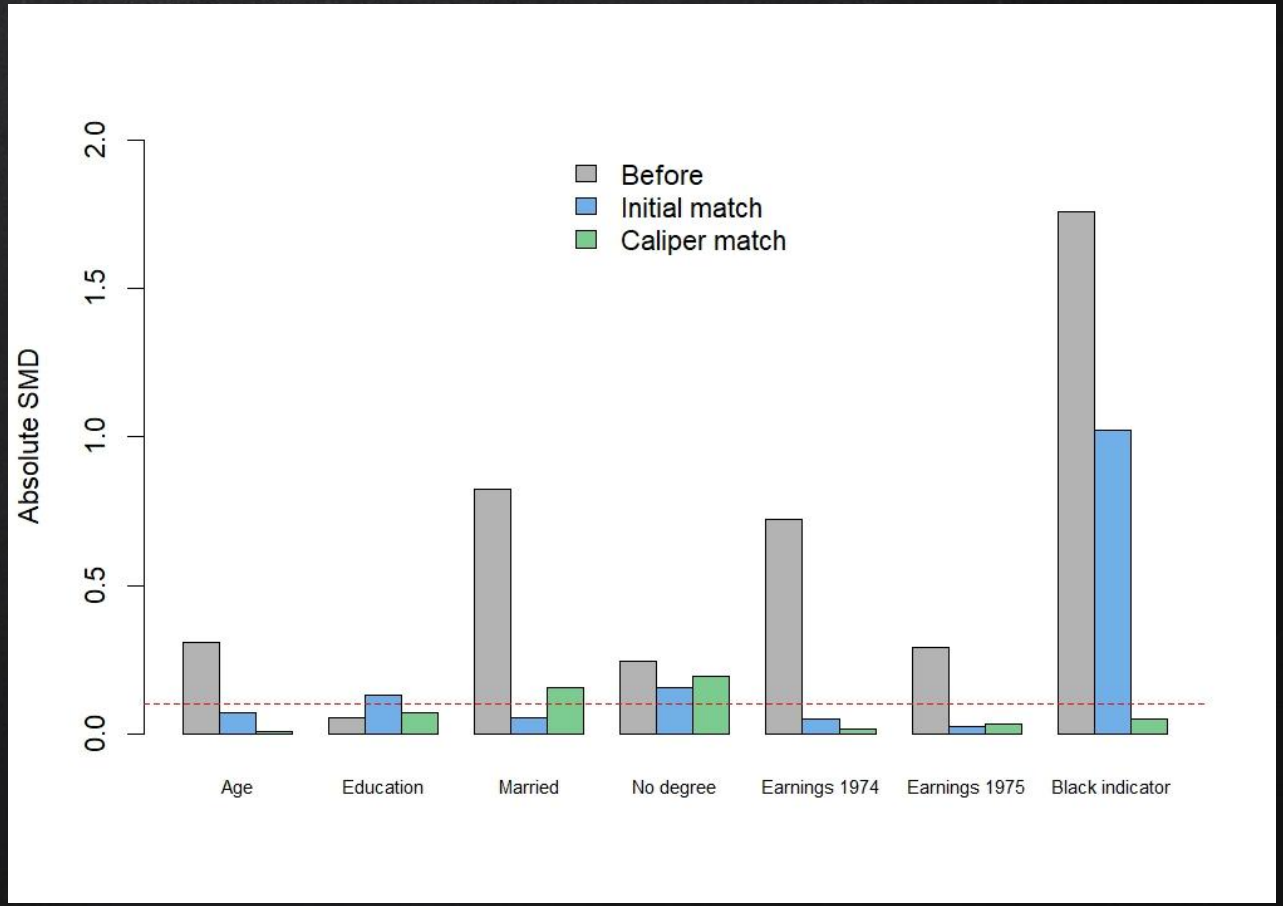
Caliper matching changes the analysis sample

Group	Initial match	0.05 caliper
Treated retained	185	113
Comparison retained	185	113
Treated unmatched	0	72

The caliper design represents retained treated participants, not automatically all 185.

Balance across matching designs

The caliper improves some imbalances, but degree status and marriage still need attention.



If balance is still poor

Revise nonlinear terms or interactions in the propensity model.

Consider exact matching on scientifically important variables.

Consider replacement, a different distance, or a different matching method.

Reassess retention and the target population after each design change.

Improve the comparison using diagnostics, not the size or sign of the estimated effect.

The matched outcome contrast

For each retained pair, compute $D_i = Y_{i,treated} - Y_{i,comparison}$.

For equally weighted one-to-one pairs,

$$\hat{\tau}_{match} = mean(D_i).$$

This equals the difference in matched-arm means.

A causal interpretation still requires adequate adjustment and support.

Pairing supplies a comparison. Assumptions justify treating it causally.

R: compute the matched point estimate

```
md <- match.data(m1)
mu1 <- with(md, weighted.mean(re78[treat == 1],
                              weights[treat == 1]))
mu0 <- with(md, weighted.mean(re78[treat == 0],
                              weights[treat == 0]))
matched_difference <- mu1 - mu0
matched_difference
```

Keep the point estimate separate from the inference step.

Results depend on design and population

Analysis	Observed earnings contrast	Population represented
Crude full-sample	About −$635	Original observed groups
Initial nearest match	About +$894	All treated participants
0.05-caliper match	About +$1,433	113 retained treated participants

The change in estimates is not proof that the final design has removed confounding.

Uncertainty and remaining assumptions

Matched observations require an analysis that respects the matching structure.

Control reuse and unequal weights require additional care.

Standard resampling is not universally valid for nearest-neighbor estimators.

No matching method removes unmeasured confounding or absent support.

A credible report includes uncertainty, balance, retention, and causal limitations.

Matching: Summary

Matching and g-computation offer different ways to use a valid adjustment set.

Scores organize comparisons, while covariate diagnostics assess the resulting design.

Calipers can improve comparability while changing the represented population.

Outcome estimates become meaningful only after design and assumption checks.

Next: weighting lets us change contributions rather than selecting pairs.

IPW: changing each person’s contribution

Matching selects comparison participants.

Weighting assigns a numerical contribution to each observed participant.

The goal is to align covariate distributions for a chosen causal target.

We continue with training, baseline X, and 1978 earnings.

IPW uses treatment probabilities to construct the comparison.

Why inverse probabilities?

A treated person with $e(X) = 0.20$ represents a relatively rare assignment.

Giving that person weight $\frac{1}{0.20} = 5$ compensates for the selection probability.

A treated person with $e(X) = 0.80$ receives weight 1.25.

The same reasoning applies to controls using $1 - e(X)$.

Less likely observed assignments receive larger ATE weights.

ATE weights

Observed treatment	Probability of that assignment	ATE weight
$T = 1$	$e(X)$	$1/e(X)$
$T = 0$	$1 - e(X)$	$1/\{1 - e(X)\}$

$$w_i = \frac{T_i}{\hat{e}_i} + \frac{1 - T_i}{1 - \hat{e}_i}$$

Actual weights for job-training participants

Participant	T	Estimated score	ATE weight
NSW1	1	0.639	1.57
NSW11	1	0.689	1.45
PSID1	0	0.026	1.03
PSID16	0	0.056	1.06

For an untreated participant, use 1/(1 − score), not 1/score.

Why weighting balances the population

The observed treated covariate density is proportional to $e(x)f(x)$.

Multiplying by $1/e(x)$ leaves a density proportional to $f(x)$.

The control density is proportional to $\{1 - e(x)\}f(x)$.

Multiplying by $1/\{1 - e(x)\}$ also leaves $f(x)$.

With the true probabilities, both ATE-weighted arms target the same $f(x)$.

Why IPW identifies a treatment mean

$$E[TY/e(X) \mid X] = E[Y \mid T = 1, X].$$

Because $E[TY \mid X] = e(X)E[Y \mid T = 1, X]$.

Averaging over X gives $E[TY/e(X)] = E[Y(1)]$.

The control expression identifies $E[Y(0)]$ in the same way.

Exchangeability, consistency, and positivity justify the causal equality.

Weighted mean estimators

Horvitz-Thompson: $\hat{\mu}_1 = n^{-1} \Sigma T_i Y_i / \hat{e}_i$.

Normalized mean: $\hat{\mu}_1 = \Sigma T_i Y_i / \hat{e}_i \div \Sigma T_i / \hat{e}_i$.

Compute the corresponding control mean using $1 - \hat{e}_i$.

The effect estimate is $\hat{\tau} = \hat{\mu}_1 - \hat{\mu}_2$.

Our real-data calculations use normalized weighted means.

R: estimate the assignment model

```
library(MatchIt)
data("lalonde", package = "MatchIt")
d <- lalonde
ps_fit <- glm(treat ~ age + educ + race + married +
              nodegree + re74 + re75,
              family = binomial(), data = d)
d$ps <- predict(ps_fit, type = "response")
stopifnot(all(d$ps > 0 & d$ps < 1))
```

This check prevents invalid denominators. It does not verify population positivity.

R: ATE weights and point estimate

```
d$w_ate <- ifelse(d$treat == 1,
                  1 / d$ps, 1 / (1 - d$ps))
mu1 <- with(d, weighted.mean(re78[treat == 1],
                             w_ate[treat == 1]))
mu0 <- with(d, weighted.mean(re78[treat == 0],
                             w_ate[treat == 0]))
ate_ipw <- mu1 - mu0
c(mu1 = mu1, mu0 = mu0, ATE = ate_ipw)
```

The target is the overall population represented by these study data.

ATE weighting: real-data results

Quantity	Estimate
Weighted training mean	$6,648
Weighted comparison mean	$6,423
Weighted difference	About +$225

This is an estimate under the proposed model and assumptions, not a validated causal conclusion.

ATT weighting

Observed treatment	ATT weight	Purpose
$T = 1$	1	Retain the treated population
$T = 0$	$e(X)/\{1 - e(X)\}$	Make controls resemble treated units

ATT weights target the treated covariate distribution, not the full-population ATE.

R: ATT weights and point estimate

```
d$w_att <- ifelse(d$treat == 1,
                  1, d$ps / (1 - d$ps))
mu1_att <- with(d, weighted.mean(re78[treat == 1],
                                 w_att[treat == 1]))
mu0_att <- with(d, weighted.mean(re78[treat == 0],
                                 w_att[treat == 0]))
att_ipw <- mu1_att - mu0_att
c(mu1 = mu1_att, mu0 = mu0_att, ATT = att_ipw)
```

Here the treated mean stays $6,349. The weighted control mean is about $5,135.

ATE and ATT answer different questions

Target	Weighting estimate	Whose effect?
ATE	About +$225	Overall study population
ATT	About +$1,214	Treated participants

Different targets can produce different effects, even with the same data and propensity model.

Stabilized ATE weights

Treated: $P(T = 1)/e(X)$.

Control: $P(T = 0)/\{1 - e(X)\}$.

Use the observed treatment fraction to estimate the numerators.

The numerators multiply all weights within an arm by the same constant.

Normalized arm means are unchanged. Stabilization does not fix poor overlap.

Extreme weights and influence

For a treated person with $\hat{e} = 0.025$, the ATE weight is about 40.

In this real dataset, the maximum treated ATE weight is 40.08.

A small number of low-score treated people can drive the weighted treated mean.

Check the weight distribution separately in each arm.

Retaining everyone does not mean everyone contributes equally.

Effective sample size

$$ESS = (\Sigma w_i)^2 / \Sigma w_i^2.$$

Equal weights give ESS equal to the observed sample size.

A few dominant weights lower ESS.

Compute ESS within each treatment arm.

ESS describes weight concentration, not a new count of independent participants.

Weight diagnostics in the real data

Design	Treated n	Treated ESS	Control n	Control ESS
ATE weights	185	58.3	429	329.0
ATT weights	185	185.0	429	99.8

ATE weights concentrate influence among treated units. ATT weights concentrate it among controls.

R: inspect weights and ESS

```
ess <- function(w) sum(w)^2 / sum(w^2)
q <- c(0, .5, .9, .95, .99, 1)
tapply(d$w_ate, d$treat, quantile, probs = q)
tapply(d$w_ate, d$treat, ess)
tapply(d$w_att, d$treat, ess)
# Largest ATE weights and their underlying scores
ix <- order(d$w_ate, decreasing = TRUE)[1:5]
d[ix, c("treat", "ps", "w_ate", "age", "re74", "re75")]
```

Inspect the people behind the largest contributions.

R: calculate the chart's balance values

```
X <- data.frame(age=d$age, educ=d$educ, married=d$married,
 nodegree=d$nodegree, re74=d$re74, re75=d$re75,
 black=as.numeric(d$race == "black"))
A <- d$treat
sref <- sapply(X[A == 1, ], sd)
smd <- function(w) sapply(seq_along(X), function(j) {
 x <- X[[j]]
 abs(weighted.mean(x[A == 1], w[A == 1]) -
     weighted.mean(x[A == 0], w[A == 0])) / sref[j]
})
```

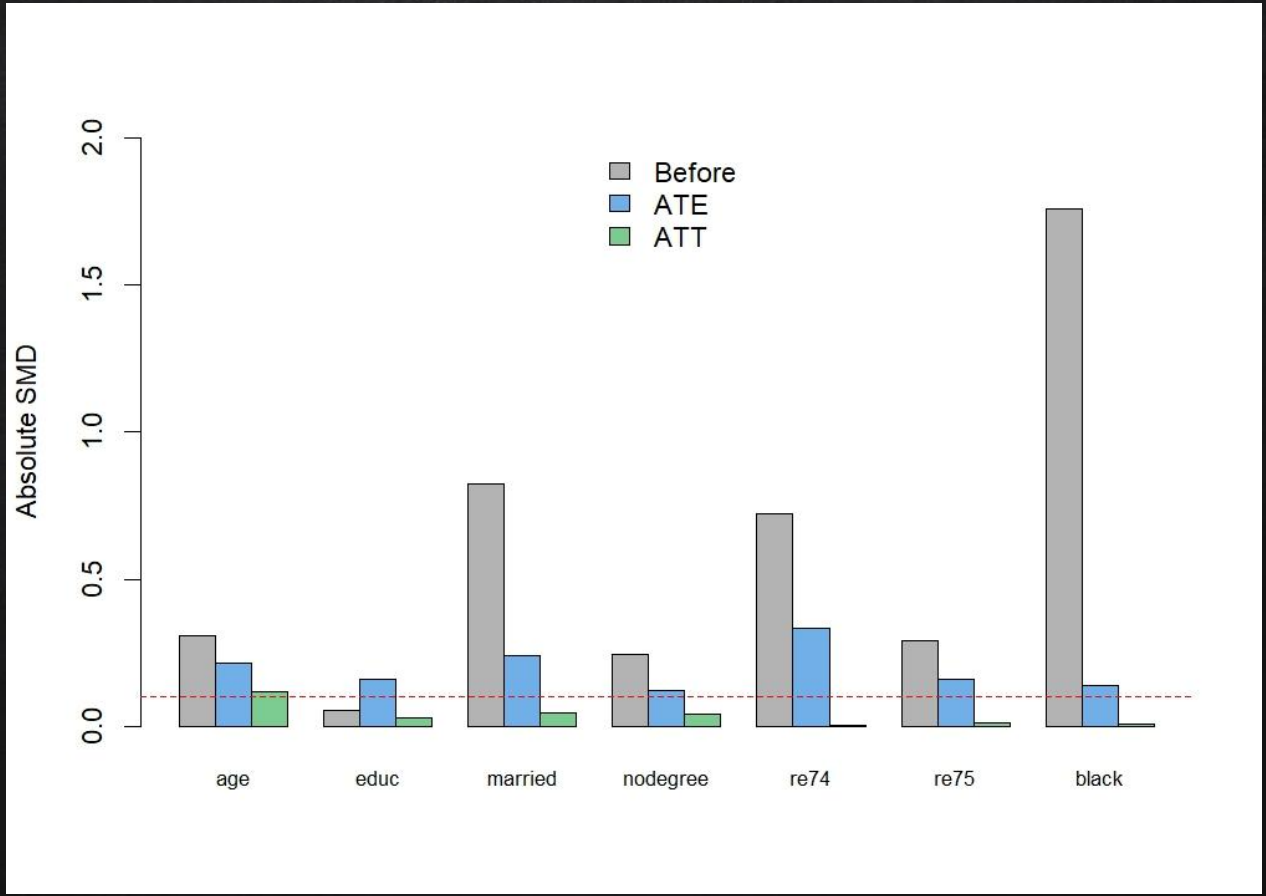
Use one fixed denominator before and after weighting.

R: reproduce the balance plot

```
balance <- rbind(
 Before = smd(rep(1, nrow(d))),
 ATE = smd(d$w_ate), ATT = smd(d$w_att))
barplot(balance, beside = TRUE,
 names.arg = names(X), cex.names = .75,
 col = c("grey70", "#71AFE8", "#7CCB91"),
 ylab = "Absolute SMD", ylim = c(0, 2),
 legend.text = rownames(balance),
 args.legend = list(x = "top", bty = "n"))
abline(h = .10, lty = 2, col = "red")
```

The dashed 0.10 line is a guideline, not a test of exchangeability.

Balance after weighting



Poor overlap: possible responses

Reconsider whether the full-population ATE is supported.

Improve the propensity specification if important modeled terms are missing.

Restriction changes the target population. Weight capping can introduce bias.

Recheck balance and sensitivity after any change.

A numerical weight modification cannot restore missing support.

Uncertainty and the next estimator

Use inference that accounts for the weighting analysis and any clustering.

A suitable refitting bootstrap can include propensity-model estimation for smooth IPW.

Sampling intervals do not quantify unmeasured confounding.

Next, AIPW combines outcome predictions with weighted residual corrections.

IPW depends on the treatment model. AIPW adds an outcome-model safeguard.

AIPW: why combine the two approaches?

G-computation predicts outcomes under training and no training.

IPW uses treatment probabilities to reconstruct comparable populations.

Each approach depends on a different statistical model.

AIPW combines predictions with weighted observed prediction errors.

Our target in this section is the same full-study-population ATE.

The two models

Model	Definition	In the job-training data
Outcome	$m_t(x) = E[Y \mid T = t, X = x]$	Predict 1978 earnings
Treatment	$e(x) = P(T = 1 \mid X = x)$	Predict training membership

Both models use the baseline adjustment set justified by the causal question.

Start with outcome predictions

For each person, predict $\widehat{m}_1(X_i)$ under training.

Also predict $\widehat{m}_0(X_\mathrm{i})$ under no training.

G-computation averages these predictions over all participants.

If the outcome model is wrong, prediction errors may systematically affect the mean.

AIPW starts with the same predictions, then adds a correction.

Observed residuals supply the correction

For a treated person, the observed training residual is $Y_i - \hat{m}_1(X_i)$.

For a control, the observed no-training residual is $Y_i - \hat{m}_0(X_i)$.

We only observe the residual in the arm received.

Inverse probabilities adjust which people supply those residuals.

The correction uses observed outcomes without inventing missing residuals.

The augmented training mean

$$\psi_{1\mathrm{i}} = \widehat{m}_1(X_i) + T_i\{Y_i - \widehat{m}_1(X_i)\}/\hat{e}_i.$$

Every person contributes a predicted training outcome.

Only treated people contribute a training residual correction.

$$\hat{\mu}_{1,AIPW} = n^{-1}\Sigma_i\, \psi_{1i}.$$

Prediction + inverse-probability-weighted residual

The augmented no-training mean

$$\psi_{1\mathrm{i}} = \widehat{m}_0(X_i) + (1 - T_i)\{Y_i - \widehat{m}_0(X_i)\}/(1 - \hat{e}_i).$$

Every person contributes a predicted no-training outcome.

Only controls contribute a no-training residual correction.

$$\hat{\mu}_{0,AIPW} = n^{-1}\Sigma_i \, \psi_{0i}.$$

Both means average over the same target population.

AIPW ATE

$$\hat{\tau}_{AIPW} = n^{-1}\Sigma_i [\psi_{1i} - \psi_{0i}].$$

Prediction contrast: $\hat{m}_1(X_i) - \hat{m}_0(X_i)$.

Add the treated residual correction $T_i\{Y_i - \hat{m}_1(X_i)\}/\hat{e}_i$.

Subtract the control residual correction $(1 - T_i)\{Y_i - \hat{m}_0(X_i)\}/(1 - \hat{e}_i)$.

The control correction enters with a minus sign.

Worked real participant: NSW1

Quantity	Value
Treatment and observed earnings	T=1, Y=$9,930
Predicted training earnings	$5,964
Estimated treatment probability	0.639
Augmented training contribution	5964 + (9930 − 5964)/0.639 ≈ $12,173

Its control contribution is the prediction $\hat{m}_0(X) \approx \$4,416$.

Worked real participant: PSID1

Quantity	Value
Treatment and observed earnings	T=0, Y=$25,565
Predicted no-training earnings	$15,935
Estimated treatment probability	0.026
Augmented control contribution	15935 + (25565 − 15935)/(1 − 0.026) ≈ $25,823

Its training contribution is the prediction $\hat{m}_1(X) \approx \$17,483.$

Why it works: a correct outcome model

Suppose $\widehat{m}_1(X)$ converges to the true $m_1(X)$.

Then $E[Y - m_1(X) \mid T = 1, X] = 0$.

The weighted residual correction has conditional expectation zero.

The average prediction identifies $E[Y(1)]$.

A correct outcome model makes the correction mean-zero.

Why it works: a correct propensity model

Suppose $\hat{e}(X)$ converges to the true $e(X)$.

Let $\tilde{m}_1(X)$ be a possibly wrong outcome-model limit.

$$E[\tilde{m}_1(X) + T\{Y - \tilde{m}_1(X)\}/e(X) \mid X]$$

$$= \tilde{m}_1(X) + m_1(X) - \tilde{m}_1(X) = m_1(X).$$

Correct assignment probabilities remove the limiting prediction error.

Two errors interact in the remaining bias

Let g(X) be the propensity-model limit.

Let $\tilde{m}_1(X)$ be the outcome-model limit.

Bias of the augmented training mean:

$$E\left[\frac{\{g(X) - e(X)\}\{\tilde{m}_1(X) - m_1(X)\}}{g(X)}\right].$$

If either error is zero, this bias term vanishes.

What double robustness does not fix

An unmeasured confounder can invalidate both model-based causal interpretations.

Missing support can prevent identification for part of the target population.

Scores near 0 or 1 can create extreme residual corrections.

Two misspecified models can still yield a biased estimate.

Double robustness concerns statistical models, not a replacement for causal assumptions.

R: fit the outcome and treatment models

```
library(MatchIt)
data("lalonde", package = "MatchIt")
d <- lalonde
f_x <- ~ age + educ + race + married + nodegree + re74 + re75
fit_y <- lm(update(f_x, re78 ~ treat + .), data = d)
fit_e <- glm(update(f_x, treat ~ .),
             family = binomial(), data = d)
m1 <- predict(fit_y, transform(d, treat = 1))
m0 <- predict(fit_y, transform(d, treat = 0))
e <- predict(fit_e, type = "response")
```

The outcome model predicts earnings. The treatment model predicts assignment.

R: calculate augmented contributions

```
A <- d$treat
Y <- d$re78
stopifnot(all(e > 0 & e < 1))
psi1 <- m1 + A * (Y - m1) / e
psi0 <- m0 + (1 - A) * (Y - m0) / (1 - e)
psi <- psi1 - psi0
ate_aipw <- mean(psi)
c(mu1 = mean(psi1), mu0 = mean(psi0), ATE = ate_aipw)
```

Prediction errors use observed minus predicted outcomes.

R: compare the same ATE target

```
ate_g <- mean(m1 - m0)
w <- ifelse(A == 1, 1 / e, 1 / (1 - e))
ate_ipw <- weighted.mean(Y[A == 1], w[A == 1]) -
           weighted.mean(Y[A == 0], w[A == 0])
c(G_computation = ate_g,
  IPW_normalized = ate_ipw,
  AIPW = ate_aipw)
```

All three calculations target the overall study-population ATE.

Real-data estimates

Method	ATE estimate	Main statistical dependence
G-computation	About +$1,548	Outcome model
Normalized IPW	About +$225	Propensity model
AIPW	About +$887	Either model for consistency

AIPW's position between these estimates does not prove it is closest to the truth.

What changed relative to g-computation?

Average predicted training earnings $\approx \$7{,}875$.

Training residual correction $\approx -\$557$, giving $\$7{,}318$.

Average predicted control earnings $\approx \$6{,}326$.

Control residual correction $\approx +\$104$, giving $\$6{,}431$.

AIPW difference: $7318 - 6431 \approx \$887$.

Uncertainty: what the simple score SE requires

When both nuisance functions are consistently estimated and suitable regularity holds:

$\hat{\phi}_i = \psi_i - \hat{\tau}_{AIPW}$ approximates an influence contribution.

$SE \approx sd(\psi_i)/\sqrt{n}$, with a normal confidence interval.

Consistency with one correct model alone does not justify this SE universally.

Use a variance procedure justified for the fitted models and target.

R: a refitting bootstrap function

```
estimate_aipw <- function(z) {
 fy <- lm(re78 ~ treat + age + educ + race + married +
          nodegree + re74 + re75, data = z)
 fe <- glm(treat ~ age + educ + race + married +
           nodegree + re74 + re75, binomial(), data = z)
 p <- predict(fe, type = "response")
 y1 <- predict(fy, transform(z, treat = 1))
 y0 <- predict(fy, transform(z, treat = 0))
 A <- z$treat; Y <- z$re78
 mean(y1 - y0 + A*(Y-y1)/p - (1-A)*(Y-y0)/(1-p))
}
```

Each resample refits both models before calculating AIPW.

R: bootstrap uncertainty

```
set.seed(7900)
B <- 500
boot_est <- replicate(B, {
 idx <- sample.int(nrow(d), replace = TRUE)
 estimate_aipw(d[idx, ])
})
stopifnot(all(is.finite(boot_est)))
boot_se <- sd(boot_est)
ci <- ate_aipw + c(-1, 1) * qnorm(.975) * boot_se
c(ATE = ate_aipw, SE = boot_se, lower = ci[1], upper = ci[2])
```

Inspect resampling stability and convergence warnings before reporting the interval.

Flexible models and cross-fitting

Flexible models can estimate nonlinear outcome and assignment functions.

Fit nuisance models outside each held-out fold.

Compute each held-out participant's score using those excluded-data fits.

Average all held-out scores and assess the required convergence conditions.

Cross-fitting limits overfitting effects. It does not fix confounding or support.

AIPW: the main lesson

Outcome predictions supply the treatment and control means.

Weighted factual residuals correct systematic prediction errors.

Either correctly specified nuisance side can protect consistency.

Causal assumptions, overlap, and suitable uncertainty quantification remain essential.

The final report should state the target, models, diagnostics, estimate, and limitations.